

Wireshark

Wireshark is a network packet analyzer used for packet capture and network analysis. It helps reveal data at microscopic level to detect any malicious activities, any unusual behaviour, deep packet inspection, vulnerability analysis, etc.

Frames:

Top Frame: The top frame in Wireshark consists of Number, Time, Src(Source IP), Dst(Destination IP), protocol type, length of packet, info

Middle Frame: The middle frame contains detail information like Frame with bytes information, IP version, Src IP, Dst IP, Ethernet, IPv4/IPv6, TCP/UDP, Src port, Dst port, Seq, Length

Bottom Frame: This frame contains data in hexadecimal and ASCII format.

The screenshot displays the Wireshark interface with the following components:

- Menu Bar:** File, Edit, View, Go, Capture, Analyze, Statistics, Telephony, Wireless, Tools, Help.
- Toolbar:** Icons for file operations, capture, analysis, and display.
- Filter Bar:** "Apply a display filter ... <Ctrl-/>"
- Packet List:** A table with columns: No., Time, Source, Destination, Protocol, Length, Info. It lists 18 packets, with packet 11204 selected.
- Packet Details:** A tree view showing the structure of the selected packet (11204):
 - > Frame 11204: Packet, 4954 bytes on wire (39632 bits), 4954 bytes captured (39632 bits) on interface \Device\NPF...
 - > Ethernet II, Src: VantivaConne_ed:11:bf (70:13:01:ed:11:bf), Dst: Intel_31:65:55 (50:e0:85:31:65:55)
 - > Internet Protocol Version 6, Src: 2607:f8b0:4023:1804::13, Dst: 2607:fea8:1bea:f200::ed1b:05:2aa9:4a29
 - > Transmission Control Protocol, Src Port: 443, Dst Port: 50424, Seq: 4640745, Ack: 384671, Len: 4880
 - > Transport Layer Security
- Packet Bytes:** A hex/ASCII view of the packet data, showing hexadecimal values on the left and their corresponding ASCII characters on the right.
- Status Bar:** "Packets: 11208 - Dropped: 0 (0.0%) Profile: Default"

Sorting and Searching:

To sort the traffic according to protocol, click on protocol in the top frame and it will sort in ascending order.

The Wireshark interface displays a list of network packets. The top pane shows the packet list, sorted by protocol. The selected packet is a DNS query (No. 673) from source 2607:fea8:1bea:f200:: to destination 2607:f798:18:10:0:6::. The middle pane shows the packet details, including Ethernet II, Internet Protocol Version 4, and Transport Layer Security. The bottom pane shows the raw packet data in hexadecimal and ASCII.

No.	Time	Source	Destination	Protocol	Length	Info
5	1.181914	Espressif_24:c2:29	Broadcast	ARP	42	ARP Announcement for 10.0.0.171
30	7.446320	VantivaConne_ed:11:...	Intel_31:65:55	ARP	42	Who has 10.0.0.226? Tell 10.0.0.1
31	7.446383	Intel_31:65:55	VantivaConne_ed:11:...	ARP	42	10.0.0.226 is at 50:e0:85:31:65:55
33	7.635482	GlazeroTechn_05:12:...	Broadcast	ARP	42	ARP Announcement for 10.0.0.139
34	7.635482	GlazeroTechn_05:12:...	Broadcast	ARP	42	ARP Announcement for 10.0.0.139
36	7.739257	GlazeroTechn_05:12:...	Broadcast	ARP	42	ARP Announcement for 10.0.0.139
37	7.840403	GlazeroTechn_05:12:...	Broadcast	ARP	42	ARP Announcement for 10.0.0.139
678	12.549302	GlazeroTechn_05:1c:...	Broadcast	ARP	42	ARP Announcement for 10.0.0.250
672	12.656898	GlazeroTechn_05:1c:...	Broadcast	ARP	42	ARP Announcement for 10.0.0.250
673	12.753322	GlazeroTechn_05:1c:...	Broadcast	ARP	42	ARP Announcement for 10.0.0.250
1518	20.025773	Espressif_ec:ba:5d	Broadcast	ARP	42	ARP Announcement for 10.0.0.6
2055	21.150732	Espressif_24:c2:29	Broadcast	ARP	42	ARP Announcement for 10.0.0.171
89	11.340522	2607:fea8:1bea:f200::	2607:f798:18:10:0:6::	DNS	116	Standard query 0x8923 HTTPS rr2----sn-gvbxgn-t34e.googlevideo.com
90	11.340706	2607:fea8:1bea:f200::	2607:f798:18:10:0:6::	DNS	116	Standard query 0x9600 AAAA rr2----sn-gvbxgn-t34e.googlevideo.com
91	11.340818	2607:fea8:1bea:f200::	2607:f798:18:10:0:6::	DNS	116	Standard query 0xd9c5 A rr2----sn-gvbxgn-t34e.googlevideo.com
92	11.358340	2607:f798:18:10:0:6::	2607:fea8:1bea:f200::	DNS	186	Standard query response 0x8923 HTTPS rr2----sn-gvbxgn-t34e.googlevideo.com CNAME rr2.sn-gvbxgn-t34e.googlevideo.com HTTPS
93	11.365359	2607:f798:18:10:0:6::	2607:fea8:1bea:f200::	DNS	180	Standard query response 0xd9c5 A rr2----sn-gvbxgn-t34e.googlevideo.com CNAME rr2.sn-gvbxgn-t34e.googlevideo.com A 209.148.199.13
94	11.365731	2607:f798:18:10:0:6::	2607:fea8:1bea:f200::	DNS	192	Standard query response 0x9600 AAAA rr2----sn-gvbxgn-t34e.googlevideo.com CNAME rr2.sn-gvbxgn-t34e.googlevideo.com AAAA 2607:f7...
781	15.351027	2607:fea8:1bea:f200::	2607:f798:18:10:0:6::	DNS	94	Standard query 0xc8d2 HTTPS lh3.google.com
782	15.351275	2607:fea8:1bea:f200::	2607:f798:18:10:0:6::	DNS	94	Standard query 0xb008 AAAA lh3.google.com
783	15.351482	2607:fea8:1bea:f200::	2607:f798:18:10:0:6::	DNS	94	Standard query 0xb008 AAAA lh3.google.com
788	15.376757	2607:f798:18:10:0:6::	2607:fea8:1bea:f200::	DNS	220	Standard query response 0xb008 A lh3.google.com CNAME lh2.1.google.com A 142.250.139.139 A 142.250.139.100 A 142.250.139.138 A...
789	15.376757	2607:f798:18:10:0:6::	2607:fea8:1bea:f200::	DNS	236	Standard query response 0xb008 AAAA lh3.google.com CNAME lh2.1.google.com AAAA 2607:f8b0:4023:1804::8a AAAA 2607:f8b0:4023:180...

To search by string or by hexadecimal value, go to edit and select find packet. After that choose the type you want and write the string text in the box and click find.

The Wireshark interface displays the Find packet dialog box. The search criteria is set to 'String' and 'Handshake'. The search results show a list of packets, including a QUIC IETF packet. The bottom pane shows the raw packet data in hexadecimal and ASCII.

No.	Time	Source	Destination	Protocol	Length	Info
68	9.754360	2607:f8b0:4023:1807::	2607:fea8:1bea:f200::	QUIC	264	Protected Payload (KP0)
69	9.754733	2607:fea8:1bea:f200::	2607:f8b0:4023:1807::	QUIC	97	Protected Payload (KP0), DCID=ecaf6d6f90d13b2c
70	9.755959	2607:f8b0:4023:1807::	2607:fea8:1bea:f200::	QUIC	83	Protected Payload (KP0)
71	9.756449	2607:fea8:1bea:f200::	2607:f8b0:4023:1807::	QUIC	97	Protected Payload (KP0), DCID=ecaf6d6f90d13b2c
72	9.775598	2607:f8b0:4023:1807::	2607:fea8:1bea:f200::	QUIC	85	Protected Payload (KP0)
95	11.367022	2607:fea8:1bea:f200::	2607:f798:14:41::d	QUIC	1292	Initial, DCID=7f5ac0e9d5c0abd5, PKN: 1, PADDING, PING, PING, PING, PING, CRYPTO, CRYPTO, PADDING, PING, PING, PADDING, PING, P...
96	11.367105	2607:fea8:1bea:f200::	2607:f798:14:41::d	QUIC	1292	Initial, DCID=7f5ac0e9d5c0abd5, PKN: 2, CRYPTO, PING, PING, PING, PING, PING, PING, CRYPTO, PING, PING, PADDING, CRYPTO, PADDING, CR...
97	11.367343	2607:fea8:1bea:f200::	2607:f798:14:41::d	QUIC	143	0-RTT, DCID=7f5ac0e9d5c0abd5
98	11.389750	2607:f798:14:41::d	2607:fea8:1bea:f200::	QUIC	102	Initial, SCID=ff5ac0e9d5c0abd5, PKN: 1, ACK
99	11.392067	2607:f798:14:41::d	2607:fea8:1bea:f200::	QUIC	1292	Initial, SCID=ff5ac0e9d5c0abd5, PKN: 2, ACK, PADDING
100	11.392376	2607:f798:14:41::d	2607:fea8:1bea:f200::	QUIC	1292	Handshake, SCID=ff5ac0e9d5c0abd5
101	11.392376	2607:f798:14:41::d	2607:fea8:1bea:f200::	QUIC	1292	Handshake, SCID=ff5ac0e9d5c0abd5
102	11.392376	2607:f798:14:41::d	2607:fea8:1bea:f200::	QUIC	1292	Handshake, SCID=ff5ac0e9d5c0abd5
103	11.392376	2607:f798:14:41::d	2607:fea8:1bea:f200::	QUIC	1251	Protected Payload (KP0)
104	11.392818	2607:fea8:1bea:f200::	2607:f798:14:41::d	QUIC	101	Handshake, DCID=ff5ac0e9d5c0abd5
105	11.393537	2607:fea8:1bea:f200::	2607:f798:14:41::d	QUIC	229	Protected Payload (KP0), DCID=ff5ac0e9d5c0abd5
106	11.393796	2607:fea8:1bea:f200::	2607:f798:14:41::d	QUIC	1288	Protected Payload (KP0), DCID=ff5ac0e9d5c0abd5
107	11.393848	2607:fea8:1bea:f200::	2607:f798:14:41::d	QUIC	744	Protected Payload (KP0), DCID=ff5ac0e9d5c0abd5
108	11.413840	2607:f798:14:41::d	2607:fea8:1bea:f200::	QUIC	593	Protected Payload (KP0)
109	11.413840	2607:f798:14:41::d	2607:fea8:1bea:f200::	QUIC	178	Protected Payload (KP0)
110	11.414178	2607:f798:14:41::d	2607:fea8:1bea:f200::	QUIC	85	Protected Payload (KP0)

Filtering Traffic:

Type a filter into the Apply a display filter at the top.

To filter a traffic we can use different operators like ==(equal to), !=(Not equal to) / not eq / ne, &&(And to join to different filters), or / ||, gt(greater than) / >, lt(less than) / <, >=(greater than equal to), <=(less than equal to). For strings, we can use tcp port, udp port, tcp address, tcp source port, tcp destination port, udp source port, udp destination port, dhcp, http etc.

The whole expression can be tcp.port == 80 or tcp.port eq 80, ip.src == 10.0.0.6, tcp.srcport == 443, tcp port in {80 443 8080} this means tcp.port == 80 || tcp.port == 443 || tcp.port == 8080, !(ip.addr == 192.168.1.1).

Filtering by web traffic:

http.request.uri == <https://demo.example.com>

http.response.code == 200

http.request.method == "GET"

tcp contains "admin"

To apply IPv4 or IPv6 or anything from the 2nd frame as filter, just right click and select apply as filter.

Here are examples of how to filter a traffic according to the requirements:

Filter by source port. Here we have filtered the packets with tcp source port 443.

The image shows a Wireshark packet capture window. The filter bar at the top contains the expression `tcp.srcport == 443`. The packet list pane shows a list of captured packets, with the first packet (No. 1) being a TCP packet from 10.0.0.6 to 192.168.1.1 on port 443. The packet details pane shows the structure of the selected packet, which is a TLSv1.3 packet. The packet bytes pane shows the raw data of the selected packet, including the ClientHello and ServerHello messages.

No.	Time	Source	Destination	Protocol	Length	Info
1	1.715492	2607:f98:d02:102::...	2607:fea8:1bea:f200::...	TCP	86	443 → 52872 [ACK] Seq=1 Ack=2 Win=501 Len=0 SLE=1 SRE=2
367	9.448221	2603:1030:210:f::...	2607:fea8:1bea:f200::...	TLSv1.2	248	Application Data
1099	16.730870	2607:f98:d02:102::...	2607:fea8:1bea:f200::...	TLSv1.2	98	Application Data
1101	16.730817	2607:f98:d02:102::...	2607:fea8:1bea:f200::...	TCP	74	443 → 52872 [FIN, ACK] Seq=25 Ack=3 Win=501 Len=0
1105	16.735763	2607:f8b0:4023:1803::...	2607:fea8:1bea:f200::...	TCP	86	443 → 55091 [SYN, ACK] Seq=0 Ack=1 Win=65535 Len=0 MSS=1440 SACK_PERM WS=256
1107	16.736156	2607:f8b0:4023:1803::...	2607:fea8:1bea:f200::...	TCP	86	443 → 57235 [SYN, ACK] Seq=0 Ack=1 Win=65535 Len=0 MSS=1440 SACK_PERM WS=256
1111	16.751137	2607:f8b0:4023:1803::...	2607:fea8:1bea:f200::...	TCP	74	443 → 55091 [ACK] Seq=1 Ack=1441 Win=268032 Len=0
1112	16.751193	2607:f8b0:4023:1803::...	2607:fea8:1bea:f200::...	TCP	74	443 → 55091 [ACK] Seq=1 Ack=2245 Win=267264 Len=0
1114	16.758131	2607:f8b0:4023:1803::...	2607:fea8:1bea:f200::...	TCP	74	443 → 57235 [ACK] Seq=1 Ack=2241 Win=267264 Len=0
1115	16.779501	2607:f8b0:4023:1803::...	2607:fea8:1bea:f200::...	TCP	1294	443 → 55091 [ACK] Seq=1 Ack=2245 Win=267264 Len=1220 [TCP PDU reassembled in 1116]
1116	16.779824	2607:f8b0:4023:1803::...	2607:fea8:1bea:f200::...	TLSv1.3	318	Server Hello, Change Cipher Spec, Application Data
1119	16.785187	2607:f8b0:4023:1803::...	2607:fea8:1bea:f200::...	TCP	1294	443 → 57235 [ACK] Seq=1 Ack=2241 Win=267264 Len=1220 [TCP PDU reassembled in 1120]
1120	16.785519	2607:f8b0:4023:1803::...	2607:fea8:1bea:f200::...	TLSv1.3	316	Server Hello, Change Cipher Spec, Application Data
1123	16.802952	2607:f8b0:4023:1803::...	2607:fea8:1bea:f200::...	TLSv1.3	1034	Application Data, Application Data
1124	16.803890	2607:f8b0:4023:1803::...	2607:fea8:1bea:f200::...	TLSv1.3	1030	Application Data, Application Data
1137	16.865215	2607:f8b0:4023:1803::...	2607:fea8:1bea:f200::...	TLSv1.3	105	Application Data
1138	16.865924	2607:f8b0:4023:1803::...	2607:fea8:1bea:f200::...	TCP	74	443 → 55091 [ACK] Seq=2456 Ack=3882 Win=265984 Len=0
1141	16.873325	2607:f8b0:4023:1803::...	2607:fea8:1bea:f200::...	TCP	74	443 → 55091 [ACK] Seq=2456 Ack=6762 Win=263168 Len=0
1142	16.873325	2607:f8b0:4023:1803::...	2607:fea8:1bea:f200::...	TCP	74	443 → 55091 [ACK] Seq=2456 Ack=8027 Win=262144 Len=0
1145	16.878034	2607:f8b0:4023:1803::...	2607:fea8:1bea:f200::...	TCP	74	443 → 55091 [ACK] Seq=2456 Ack=10907 Win=259328 Len=0
1146	16.878424	2607:f8b0:4023:1803::...	2607:fea8:1bea:f200::...	TCP	74	443 → 55091 [ACK] Seq=2456 Ack=13270 Win=257024 Len=0
1149	16.917947	2607:f8b0:4023:1803::...	2607:fea8:1bea:f200::...	TLSv1.3	1294	Application Data

> Frame 1124: Packet, 1030 bytes on wire (8240 bits), 1030 bytes captured (8240 bits) on interface Device
> Ethernet II, Src: VantivaComme, ed:11:b7 (70:13:01:ed:11:b7), Dst: Intel_31:65:55 (50:e0:85:31:65:55)
> Internet Protocol Version 6, Src: 2607:f8b0:4023:1803::11, Dst: 2607:fea8:1bea:f200::ed1b:1a05:2aa9:4a29
> Transmission Control Protocol, Src Port: 443, Dst Port: 57235, Seq: 1463, Ack: 2315, Len: 956
> Transport Layer Security

0000 50 e0 85 31 65 55 70 13 01 ed 11 b7 86 dd 60 09 P...leUp...
0010 09 df 03 d0 06 78 26 07 f8 b0 40 23 18 03 00 00 ...x&...@...
0020 00 00 00 00 00 11 26 07 fe a8 1b ea f2 00 ed 1b ...&...
0030 1a 05 2a a9 4a 29 01 bb df 93 f9 e4 cd 8a 26 5e ...*...&...
0040 e4 0f 50 18 04 14 be 41 00 00 17 03 03 79 ac ...P...A...y...
0050 07 1f 71 44 a5 93 c2 19 b8 1a dc f3 85 e9 79 05 ...qD...Y...
0060 1f ac 5f 7c 6e f9 39 dc 40 ac 82 6c 9f aa 8b d5 ...|n.9 @...1...
0070 9c e8 ca a4 b6 23 ff a5 37 30 f9 19 2d 09 54 10 ...#...70...T...
0080 34 80 a2 a8 68 18 6e ee ae 0a 73 15 58 98 a5 33 4...h...s...X...3
0090 48 e4 20 aa 88 7d e4 11 13 5a f0 26 ae 6e 07 65 H...}...Z...&...n...e
00a0 23 6b 53 5e dd f2 5d cf 15 a0 1d 81 0b 9b e8 24 *kS...}...\$...
00b0 7b 73 c2 3d 91 8c 92 89 66 0b de f9 af 36 0a 32 {s...-...f...6...2
00c0 60 38 a8 55 5c 1a 0e bb 72 62 07 44 ee 0c 7d 98 'B(U)...rb(D)...
00d0 ac 43 38 c7 f7 cc 3f 70 4c d0 00 a3 08 02 be 8e -CB...7p L...
00e0 85 5e 1a d3 90 da 67 cc 87 42 4e 7d eb 19 04 82 ^...g...BN)...
00f0 f0 da 01 e1 1b df be a3 7c c6 37 6d a4 b2 d8 3d ...-...7m...==

Filtered packet by tcp source port 443 or 80.

Wi-Fi

File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help

tcp.port == 443 or tcp.port == 80

No.	Time	Source	Destination	Protocol	Length	Info
1113	16.751993	2607:f8b0:4023:1803...	2607:fea8:1bea:f200...	TCP	74	443 → 55091 [ACK] Seq=1 Ack=2245 Win=267264 Len=0
1114	16.758131	2607:f8b0:4023:1803...	2607:fea8:1bea:f200...	TCP	74	443 → 57235 [ACK] Seq=1 Ack=2241 Win=267264 Len=0
1115	16.779501	2607:f8b0:4023:1803...	2607:fea8:1bea:f200...	TCP	1294	443 → 55091 [ACK] Seq=1 Ack=2245 Win=267264 Len=1220 [TCP PDU reassembled in 1116]
1116	16.779824	2607:f8b0:4023:1803...	2607:fea8:1bea:f200...	TLSv1.3	318	Server Hello, Change Cipher Spec, Application Data
1117	16.779877	2607:fea8:1bea:f200...	2607:f8b0:4023:1803...	TCP	74	55091 → 443 [ACK] Seq=2245 Ack=1465 Win=65280 Len=0
1118	16.780355	2607:fea8:1bea:f200...	2607:f8b0:4023:1803...	TLSv1.3	148	Change Cipher Spec, Application Data
1119	16.785187	2607:f8b0:4023:1803...	2607:fea8:1bea:f200...	TCP	1294	443 → 57235 [ACK] Seq=1 Ack=2241 Win=267264 Len=1220 [TCP PDU reassembled in 1120]
1120	16.785519	2607:f8b0:4023:1803...	2607:fea8:1bea:f200...	TLSv1.3	316	Server Hello, Change Cipher Spec, Application Data
1121	16.785578	2607:fea8:1bea:f200...	2607:f8b0:4023:1803...	TCP	74	57235 → 443 [ACK] Seq=2241 Ack=1463 Win=65280 Len=0
1122	16.786250	2607:fea8:1bea:f200...	2607:f8b0:4023:1803...	TLSv1.3	148	Change Cipher Spec, Application Data
1123	16.802952	2607:f8b0:4023:1803...	2607:fea8:1bea:f200...	TLSv1.3	1034	Application Data, Application Data
1124	16.803890	2607:f8b0:4023:1803...	2607:fea8:1bea:f200...	TLSv1.3	1030	Application Data, Application Data
1127	16.843241	2607:fea8:1bea:f200...	2607:f8b0:4023:1803...	TCP	74	55091 → 443 [ACK] Seq=2319 Ack=2425 Win=64512 Len=0
1128	16.849658	2607:fea8:1bea:f200...	2607:f8b0:4023:1803...	TLSv1.3	166	Application Data
1129	16.849934	2607:fea8:1bea:f200...	2607:f8b0:4023:1803...	TLSv1.3	105	Application Data
1130	16.850159	2607:fea8:1bea:f200...	2607:f8b0:4023:1803...	TLSv1.3	5117	Application Data
1131	16.850232	2607:fea8:1bea:f200...	2607:f8b0:4023:1803...	TLSv1.3	616	Application Data
1135	16.853073	2607:fea8:1bea:f200...	2607:f8b0:4023:1803...	TCP	74	57235 → 443 [ACK] Seq=2315 Ack=2419 Win=64512 Len=0
1136	16.859128	2607:fea8:1bea:f200...	2607:f8b0:4023:1803...	TLSv1.3	5317	Application Data
1137	16.865215	2607:f8b0:4023:1803...	2607:fea8:1bea:f200...	TLSv1.3	105	Application Data
1138	16.865924	2607:f8b0:4023:1803...	2607:fea8:1bea:f200...	TCP	74	443 → 55091 [ACK] Seq=2456 Ack=3882 Win=265984 Len=0
1141	16.873325	2607:f8b0:4023:1803...	2607:fea8:1bea:f200...	TCP	74	443 → 55091 [ACK] Seq=2456 Ack=6762 Win=263168 Len=0
1142	16.873325	2607:f8b0:4023:1803...	2607:fea8:1bea:f200...	TCP	74	443 → 55091 [ACK] Seq=2456 Ack=8027 Win=262144 Len=0

> Frame 1124: Packet, 1030 bytes on wire (8240 bits), 1030 bytes captured (8240 bits) on interface \Device\NPF...
> Ethernet II, Src: VantivaConne...ed:11:bf (78:13:01:ed:11:bf), Dst: Intel_31:65:55 (50:e0:85:31:65:55)
> Internet Protocol Version 6, Src: 2607:f8b0:4023:1803::11, Dst: 2607:fea8:1bea:f200:ed1b:1a05:2aa9:4a29
> Transmission Control Protocol, Src Port: 443, Dst Port: 57235, Seq: 1463, Ack: 2315, Len: 956
> Transport Layer Security

0000 50 e0 85 31 65 55 70 13 01 ed 11 bf 06 dd 60 09 P...leUp...
0010 09 df 03 d0 06 78 26 07 f8 b0 40 23 18 03 00 00 ...x&...@...
0020 00 00 00 00 00 11 26 07 fe a8 1b ea f2 00 ed 1b ...&...
0030 1a 05 2a a9 4a 29 01 bb df 93 f9 e4 cd 8a 26 5e ...*)...&^
0040 e4 0f 50 18 04 14 be 41 00 00 17 03 03 03 79 ac ...P...A...y...
0050 07 1f 71 44 a5 93 c2 19 b8 1a dc f3 85 e9 79 05 ...qd...n-9...@...1...
0060 1f ac 5f 7c 6e f9 39 dc 40 ac 82 6c 9f aa 8b d5 ...#...70...T...
0070 9c e8 ca a4 b6 23 ff a5 37 30 f9 19 2d 09 54 10 ...h...s-X...3...
0080 34 80 a2 a8 68 18 6e ee ae 0a 73 15 98 a5 33 4...h...s-X...3...
0090 48 e4 20 aa 88 7d e4 11 13 5a f0 26 ae 6e 07 65 H...}...Z&n...e...
00a0 23 6b 53 5e d4 f2 5d cf 15 a0 1d 81 0b 9b ea 24 #K\$...}...\$...
00b0 7b 73 c2 3d 91 8c 92 89 66 0b de f9 af 36 0a 32 {s...f...6:2...
00c0 60 38 a8 55 5c 1a 0e bb 72 62 07 44 ee 0c 7d 98 '8U...rb-D...)-
00d0 ac 43 38 c7 f7 cc 3f 70 4c d0 80 a3 08 02 be 8e -C8...p L...
00e0 85 5e 1a d3 90 da 67 cc 87 42 4e 7d eb 19 04 82 ^...g...BN)...
00f0 f0 da 01 e1 1b df be a3 7c c6 37 6d a4 b2 08 3d ...-7m...=

Wireshark-Wi-Fi5WQR3.pcapng

Packets: 9168 · Displayed: 2892 (31.5%) · Dropped: 0 (0.0%)

Profile: Default

Filtered packet by udp port 6667 and the source ip address to be 10.0.0.6

Wi-Fi

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udp.port == 6667 && ip.src == 10.0.0.6

No.	Time	Source	Destination	Protocol	Length	Info
9	3.069079	10.0.0.6	255.255.255.255	UDP	214	49154 → 6667 Len=172
198	8.086066	10.0.0.6	255.255.255.255	UDP	214	49154 → 6667 Len=172
899	13.103954	10.0.0.6	255.255.255.255	UDP	214	49154 → 6667 Len=172
1454	18.018773	10.0.0.6	255.255.255.255	UDP	214	49154 → 6667 Len=172
2786	23.036504	10.0.0.6	255.255.255.255	UDP	214	49154 → 6667 Len=172
3342	28.054877	10.0.0.6	255.255.255.255	UDP	214	49154 → 6667 Len=172
5659	38.090018	10.0.0.6	255.255.255.255	UDP	214	49154 → 6667 Len=172
6065	43.107196	10.0.0.6	255.255.255.255	UDP	214	49154 → 6667 Len=172
6353	48.023053	10.0.0.6	255.255.255.255	UDP	214	49154 → 6667 Len=172
6407	53.040133	10.0.0.6	255.255.255.255	UDP	214	49154 → 6667 Len=172
7635	58.057729	10.0.0.6	255.255.255.255	UDP	214	49154 → 6667 Len=172
8751	63.075922	10.0.0.6	255.255.255.255	UDP	214	49154 → 6667 Len=172
9165	68.094227	10.0.0.6	255.255.255.255	UDP	214	49154 → 6667 Len=172

> Frame 9: Packet, 214 bytes on wire (1712 bits), 214 bytes captured (1712 bits) on interface \Device\NPF...
> Ethernet II, Src: Expressif_ec:ba:5d (60:01:94:ec:ba:5d), Dst: Broadcast (ff:ff:ff:ff:ff:ff)
> Internet Protocol Version 4, Src: 10.0.0.6, Dst: 255.255.255.255
> User Datagram Protocol, Src Port: 49154, Dst Port: 6667
> Data (172 bytes)

0000 ff ff ff ff ff 60 01 94 ec ba 5d 08 00 45 00E...
0010 00 c8 52 18 00 00 ff 11 5f 07 0a 00 06 ff ff ...R...
0020 ff ff c0 82 1a 00 00 b4 43 9a 00 00 55 aa 00 00C...U...
0030 00 00 00 00 13 00 00 00 9c 00 00 00 00 bb 02
0040 d3 88 f7 8c a5 00 a4 9a b1 60 7a 79 f2 c1 0e 48H...
0050 71 bd b2 77 f7 c8 22 bb 67 b9 f3 19 80 39 ca a4 q...w...g...9...
0060 67 9a 72 47 45 7c 57 c8 d5 d0 35 da 34 3e cf c9 g...rGE[W...5:4>...
0070 31 ab 14 07 01 63 98 c6 45 b4 6f 66 41 8e 10 fd 1...c...E...ofA...
0080 0f 5d d8 de 74 50 37 de 9f 96 6e b6 04 4f ad c7 -)...tP7...n...OM...
0090 fe 82 3e 30 79 27 fb 0a 44 b0 df ae 2c 8d 76 a0 ...>0y'...D...y...v...
00a0 6b 95 b8 42 35 80 ef 1d 6a de 2f 7a 89 7b 22 1b k...B5...j/z/{"...
00b0 62 06 4b 26 a7 3e 8e be 37 8c 7d c1 f5 42 ee dd b...K8...>7...}...B...
00c0 5c 95 3b b9 37 75 60 ae 90 df bc 45 db e2 c2 df \...;7u'...E...
00d0 ed 5a 00 00 aa 55Z...U...

Wireshark-Wi-Fi5WQR3.pcapng

Packets: 9168 · Displayed: 13 (0.1%) · Dropped: 0 (0.0%)

Profile: Default

Filtered packets with udp port number 6667. The port number can be source or destination as nothing is mentioned.

Wireshark packet capture showing filtered packets with UDP port number 6667. The packet list shows various UDP and DNS packets. The packet details pane shows the structure of a packet, including Ethernet II, Internet Protocol Version 4, and User Datagram Protocol. The packet bytes pane shows the raw data in hexadecimal and ASCII.

No.	Time	Source	Destination	Protocol	Length	Info
9	3.069079	10.0.0.6	255.255.255.255	UDP	214	49154 → 6667 Len=172
10	3.478543	10.0.0.171	255.255.255.255	UDP	230	49154 → 6667 Len=188
11	4.092556	10.0.0.240	255.255.255.255	UDP	82	9487 → 9478 Len=40
12	4.914452	2607:fea8:1bea:f200::	2607:f8b0:4023:1804::	UDP	91	49223 → 443 Len=29
13	4.916638	2607:fea8:1bea:f200::	2607:f8b0:4023:1804::	UDP	1288	49223 → 443 Len=1226
14	4.916765	2607:fea8:1bea:f200::	2607:f8b0:4023:1804::	UDP	1292	49223 → 443 Len=1230
15	4.916823	2607:fea8:1bea:f200::	2607:f8b0:4023:1804::	UDP	631	49223 → 443 Len=569
16	4.916887	2607:fea8:1bea:f200::	2607:f8b0:4023:1804::	UDP	148	49223 → 443 Len=86
17	4.931340	2607:f8b0:4023:1804::	2607:fea8:1bea:f200::	UDP	87	443 → 49223 Len=25
18	4.937938	2607:f8b0:4023:1804::	2607:fea8:1bea:f200::	UDP	87	443 → 49223 Len=25
19	4.937938	2607:f8b0:4023:1804::	2607:fea8:1bea:f200::	UDP	87	443 → 49223 Len=25
20	4.966351	2607:f8b0:4023:1804::	2607:fea8:1bea:f200::	UDP	92	443 → 49223 Len=30
21	4.992760	2607:fea8:1bea:f200::	2607:f8b0:4023:1804::	UDP	94	49223 → 443 Len=32
22	5.021647	2607:f8b0:4023:1804::	2607:fea8:1bea:f200::	UDP	544	443 → 49223 Len=482
23	5.021647	2607:f8b0:4023:1804::	2607:fea8:1bea:f200::	UDP	83	443 → 49223 Len=21
24	5.022474	2607:fea8:1bea:f200::	2607:f8b0:4023:1804::	UDP	97	49223 → 443 Len=35
25	5.027165	2607:fea8:1bea:f200::	2607:f798:18:10:0:6::	DNS	99	Standard query 0x6121 HTTPS accounts.google.com
26	5.027833	2607:fea8:1bea:f200::	2607:f798:18:10:0:6::	DNS	99	Standard query 0x29ed AAAA accounts.google.com
27	5.028390	2607:fea8:1bea:f200::	2607:f798:18:10:0:6::	DNS	99	Standard query 0x4fc2 A accounts.google.com
28	5.043504	2607:f798:18:10:0:6::	2607:fea8:1bea:f200::	DNS	127	Standard query response 0x29ed AAAA accounts.google.com AAAA 2607:f8b0:4004:c07::54
29	5.043993	2607:f798:18:10:0:6::	2607:fea8:1bea:f200::	DNS	115	Standard query response 0x4fc2 A accounts.google.com A 172.253.62.84
30	5.044354	2607:f798:18:10:0:6::	2607:fea8:1bea:f200::	DNS	149	Standard query response 0x6121 HTTPS accounts.google.com SOA ns1.google.com
31	5.047263	2607:fea8:1bea:f200::	2607:f8b0:4004:c07::	QUIC	1292	Initial, DCID=5accedb5a42a70, PKN: 1, CRYPTO, CRYPTO, CRYPTO, PING

> Frame 9: Packet, 214 bytes on wire (1712 bits), 214 bytes captured (1712 bits) on interface \Device\NPF...
> Ethernet II, Src: Esspressif_ec:ba:5d (60:01:94:ec:ba:5d), Dst: Broadcast (ff:ff:ff:ff:ff:ff)
> Internet Protocol Version 4, Src: 10.0.0.6, Dst: 255.255.255.255
> User Datagram Protocol, Src Port: 49154, Dst Port: 6667
> Data (172 bytes)

Filtered packets with udp port number 6667 with no arp traffic.

Wireshark packet capture showing filtered packets with UDP port number 6667 and no ARP traffic. The packet list shows various UDP packets. The packet details pane shows the structure of a packet, including Ethernet II, Internet Protocol Version 4, and User Datagram Protocol. The packet bytes pane shows the raw data in hexadecimal and ASCII.

No.	Time	Source	Destination	Protocol	Length	Info
10	3.478543	10.0.0.171	255.255.255.255	UDP	230	49154 → 6667 Len=188
198	8.086066	10.0.0.6	255.255.255.255	UDP	214	49154 → 6667 Len=172
899	13.103954	10.0.0.6	255.255.255.255	UDP	214	49154 → 6667 Len=172
900	13.513602	10.0.0.171	255.255.255.255	UDP	230	49154 → 6667 Len=188
1454	18.018773	10.0.0.6	255.255.255.255	UDP	214	49154 → 6667 Len=172
1535	18.529185	10.0.0.171	255.255.255.255	UDP	230	49154 → 6667 Len=188
2786	23.036504	10.0.0.6	255.255.255.255	UDP	214	49154 → 6667 Len=172
2791	23.548654	10.0.0.171	255.255.255.255	UDP	230	49154 → 6667 Len=188
3342	28.054877	10.0.0.6	255.255.255.255	UDP	214	49154 → 6667 Len=172
4288	33.479713	10.0.0.171	255.255.255.255	UDP	230	49154 → 6667 Len=188
5659	38.090018	10.0.0.6	255.255.255.255	UDP	214	49154 → 6667 Len=172
5660	38.499262	10.0.0.171	255.255.255.255	UDP	230	49154 → 6667 Len=188
6065	43.107196	10.0.0.6	255.255.255.255	UDP	214	49154 → 6667 Len=172
6066	43.515273	10.0.0.171	255.255.255.255	UDP	230	49154 → 6667 Len=188
6353	48.023053	10.0.0.6	255.255.255.255	UDP	214	49154 → 6667 Len=172
6368	48.535014	10.0.0.171	255.255.255.255	UDP	230	49154 → 6667 Len=188
6407	53.040133	10.0.0.6	255.255.255.255	UDP	214	49154 → 6667 Len=172
6416	53.550884	10.0.0.171	255.255.255.255	UDP	230	49154 → 6667 Len=188
7635	58.057729	10.0.0.6	255.255.255.255	UDP	214	49154 → 6667 Len=172
8022	58.568459	10.0.0.171	255.255.255.255	UDP	230	49154 → 6667 Len=188
8751	63.075922	10.0.0.6	255.255.255.255	UDP	214	49154 → 6667 Len=172
9165	68.094227	10.0.0.6	255.255.255.255	UDP	214	49154 → 6667 Len=172
9166	68.502883	10.0.0.171	255.255.255.255	UDP	230	49154 → 6667 Len=188

> Frame 9: Packet, 214 bytes on wire (1712 bits), 214 bytes captured (1712 bits) on interface \Device\NPF...
> Ethernet II, Src: Esspressif_ec:ba:5d (60:01:94:ec:ba:5d), Dst: Broadcast (ff:ff:ff:ff:ff:ff)
> Internet Protocol Version 4, Src: 10.0.0.6, Dst: 255.255.255.255
> User Datagram Protocol, Src Port: 49154, Dst Port: 6667
> Data (172 bytes)

To save a filter, click on + and save the filter by adding label to it. Next time when you want to use the same filter, just open click on the label. This is done with frequently used filters.

Wireshark interface showing a packet capture filter: `udp.port == 6667 and !arp`. The filter is labeled "udp and no arp". The packet list shows multiple UDP packets from 10.0.0.6 to 255.255.255.255 on port 6667. The selected packet (No. 9) is a UDP packet from 10.0.0.6 to 255.255.255.255 on port 6667, with a length of 172 bytes. The packet details pane shows the Ethernet II, Internet Protocol Version 4, User Datagram Protocol, and Data (172 bytes) fields. The packet bytes pane shows the raw data in hexadecimal and ASCII.

Filter packets by ip address. It can be both source or destination as nothing specific is mentioned.

Wireshark interface showing a packet capture filter: `ip.addr == 10.0.0.226`. The packet list shows multiple packets from 10.0.0.226 to 10.0.0.240. The selected packet (No. 7) is a TCP packet from 10.0.0.226 to 10.0.0.240 on port 8009, with a length of 54 bytes. The packet details pane shows the Ethernet II, Internet Protocol Version 4, and Transmission Control Protocol fields. The packet bytes pane shows the raw data in hexadecimal and ASCII.

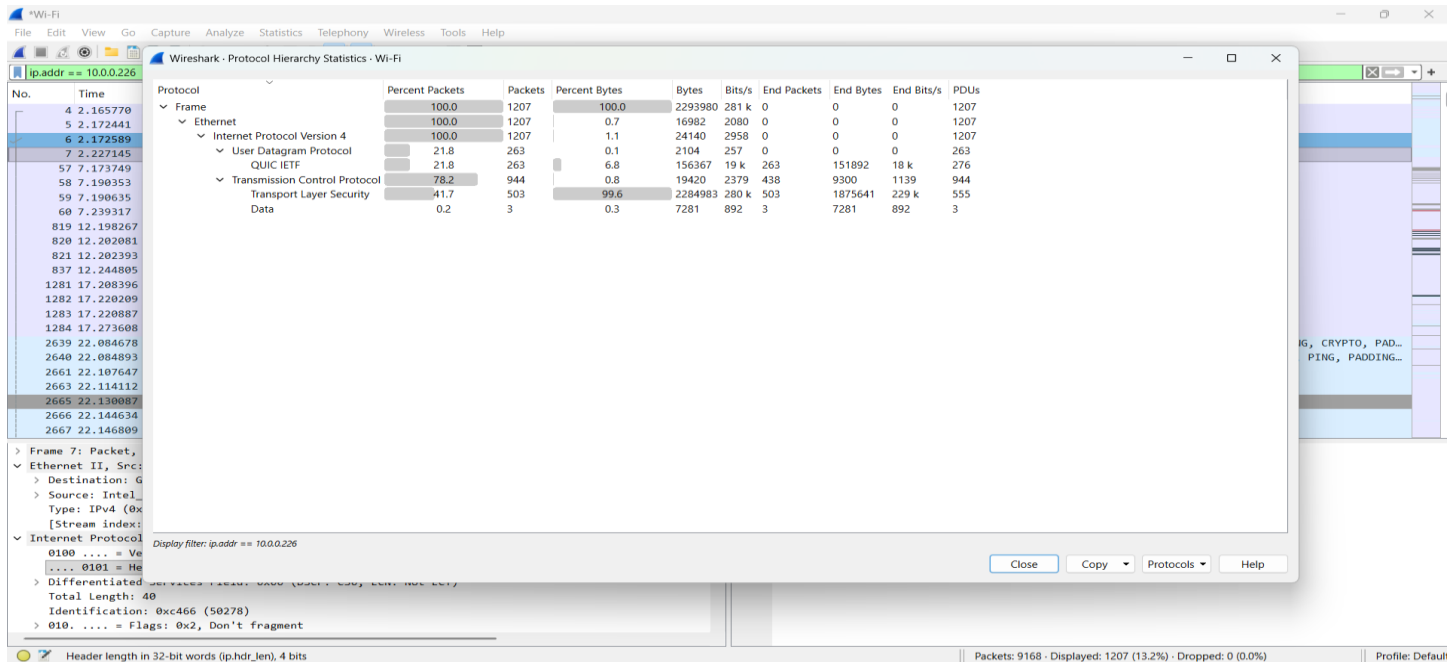
To get more info:

To dig deep and get more information, statistics is very helpful. Go to stats and select different options.

Protocol Hierarchy – Shows different protocols so it is easy to identify unexpected protocols. It also provides % of packets and bytes to analyze which protocol is using maximum bytes. To check in detail, it can also be applied as a filter by right click on the protocol.

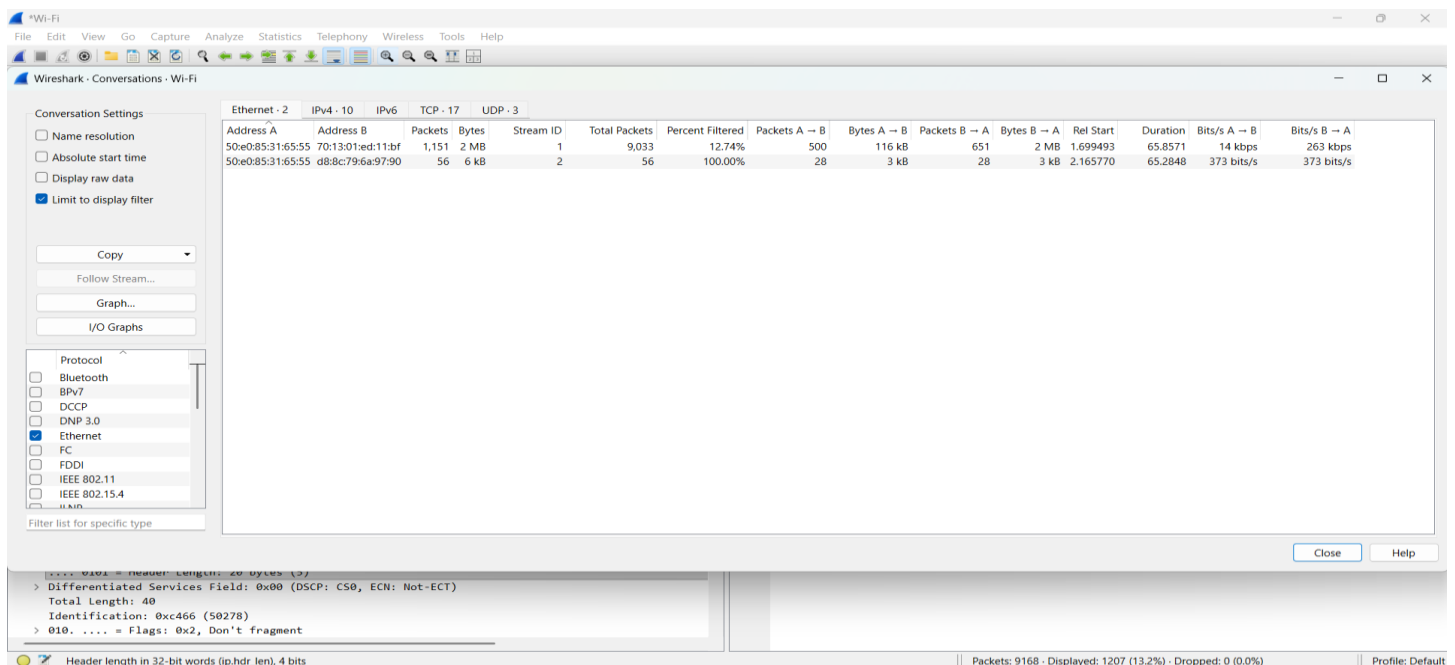
Conversations – Helpful in quickly identifying network activity, spotting unusual patterns. It shows traffic with hosts, MAC address, IPv4/ IPv6 address, volume of traffic. With the help of this we can check which IP address is sending most data/packets or sort packets to see highest traffic, TCP conversations with latency issues.

Endpoints – This is used to quickly identify key devices, reveal communication patterns, analyze traffic volume.



The image shows the Wireshark Protocol Hierarchy Statistics window. The left pane displays a list of packets with a filter 'ip.addr == 10.0.0.226'. The main pane shows a tree view of the protocol hierarchy with corresponding statistics. The bottom pane shows the details of the selected packet (Frame 7: Packet, Ethernet II, Src: 50:e0:85:31:65:55, Destination: 08:00:27:1d:33:00, Source: Intel, Type: IPv4 (0x00000001), [Stream index: 0], Internet Protocol Version 4, Total Length: 40, Identification: 0xc466 (50278), 010. = Flags: 0x2, Don't fragment).

Protocol	Percent Packets	Packets	Percent Bytes	Bytes	Bits/s	End Packets	End Bytes	End Bits/s	PDU/s
Frame	100.0	1207	100.0	2293980	281 k	0	0	0	1207
Ethernet	100.0	1207	0.7	16982	2080	0	0	0	1207
Internet Protocol Version 4	100.0	1207	1.1	24140	2958	0	0	0	1207
User Datagram Protocol	21.8	263	0.1	2104	257	0	0	0	263
QUIC IETF	21.8	263	6.8	156367	19 k	263	151892	18 k	276
Transmission Control Protocol	78.2	944	0.8	19420	2379	438	9300	1139	944
Transport Layer Security	41.7	503	99.6	2284983	280 k	503	1875641	229 k	555
Data	0.2	3	0.3	7281	892	3	7281	892	3



The image shows the Wireshark Conversations window. The left pane displays the Conversation Settings, including Name resolution, Absolute start time, Display raw data, and Limit to display filter. The main pane shows a table of conversations. The bottom pane shows the details of the selected conversation (Frame 7: Packet, Ethernet II, Src: 50:e0:85:31:65:55, Destination: 08:00:27:1d:33:00, Source: Intel, Type: IPv4 (0x00000001), [Stream index: 0], Internet Protocol Version 4, Total Length: 40, Identification: 0xc466 (50278), 010. = Flags: 0x2, Don't fragment).

Address A	Address B	Packets	Bytes	Stream ID	Total Packets	Percent Filtered	Packets A → B	Bytes A → B	Packets B → A	Bytes B → A	Rel Start	Duration	Bits/s A → B	Bits/s B → A
50:e0:85:31:65:55	70:13:01:ed:11:b1	1,151	2 MB	1	9,033	12.74%	500	116 kB	651	2 MB	1.699493	65.8571	14 kbps	263 kbps
50:e0:85:31:65:55	d8:8c:79:6a:97:90	56	6 kB	2	56	100.00%	28	3 kB	28	3 kB	2.165770	65.2848	373 bits/s	373 bits/s